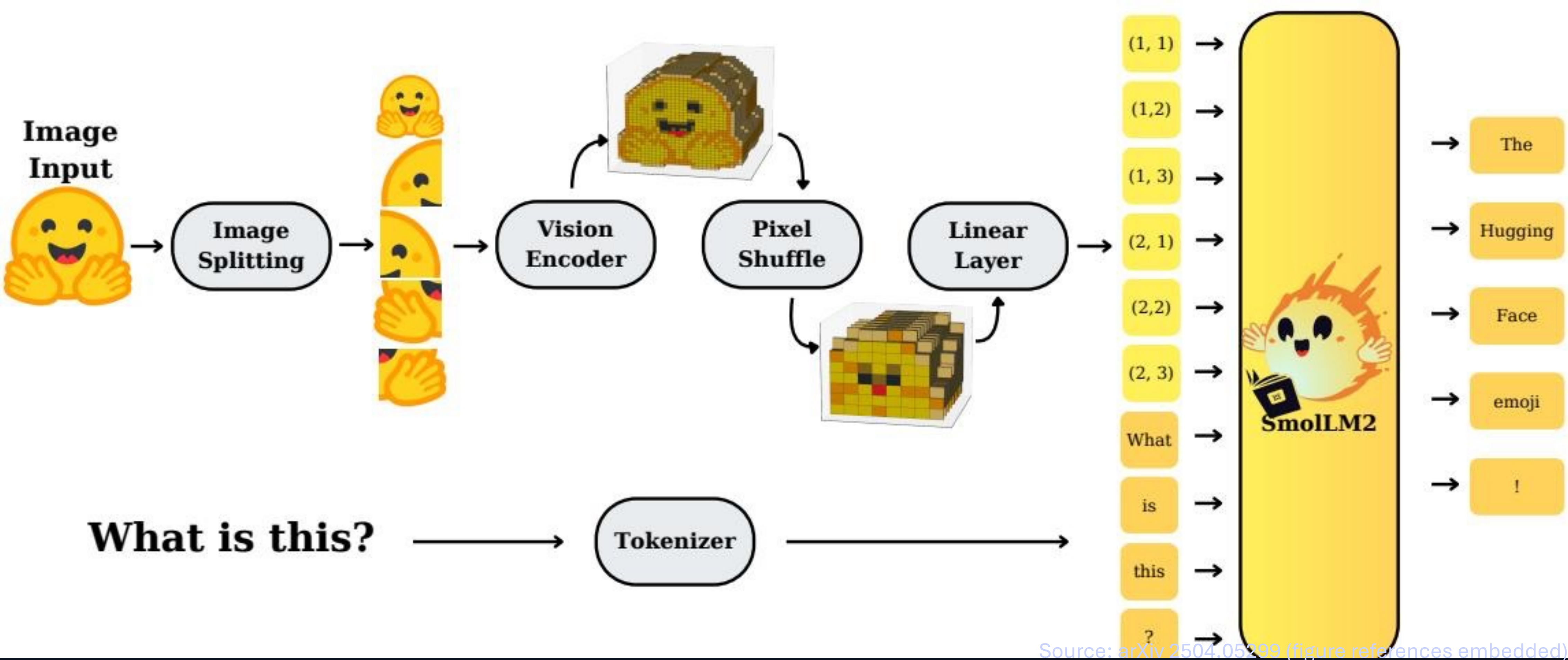
SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University • Compact VLMs for on-device inference



Source: arXiv 2504.05299 (figure references embedded)

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs push accuracy via scale — but memory dominates edge constraints.

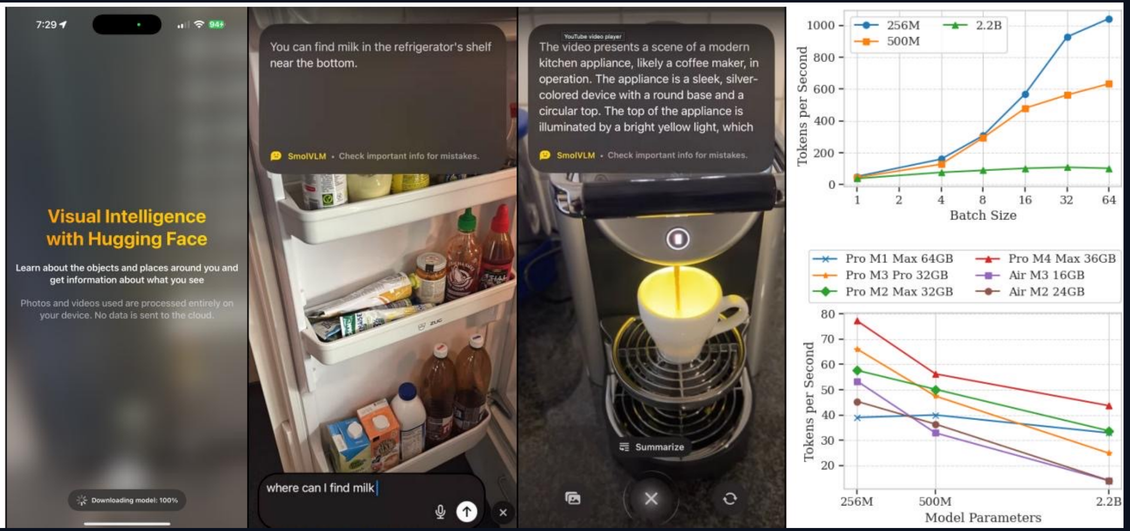
Example: Molmo-A1B-7B requires 27.7GB RAM for inference.

Goal: competitive multimodal quality with sub-1GB to single-digit GB RAM.

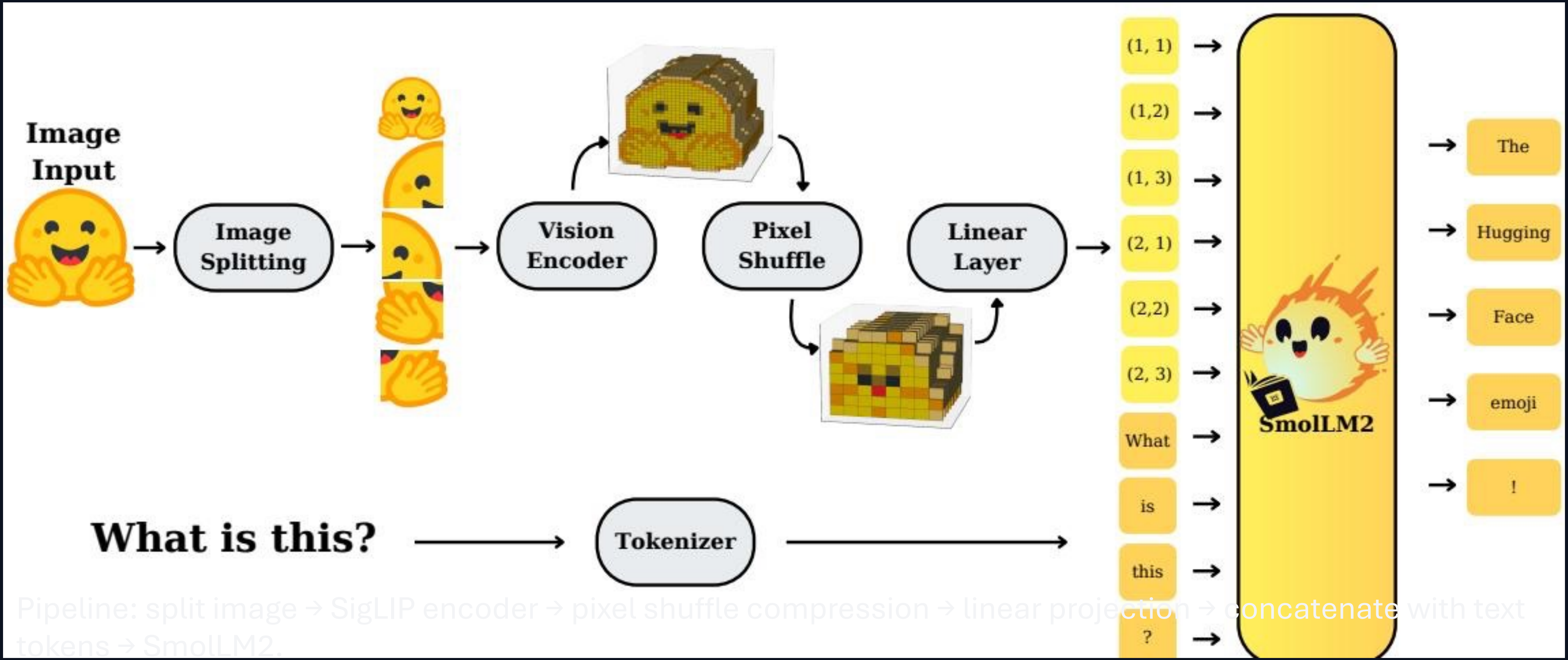
SmolVLM targets mobile/edge by redesigning compute allocation, tokenization, and context.

Memory footprint (batch=1)

SmolVLM-256M: 0.8GB
SmolVLM-2.2B: 4.9GB
Molmo-A1B-7B: 27.7GB



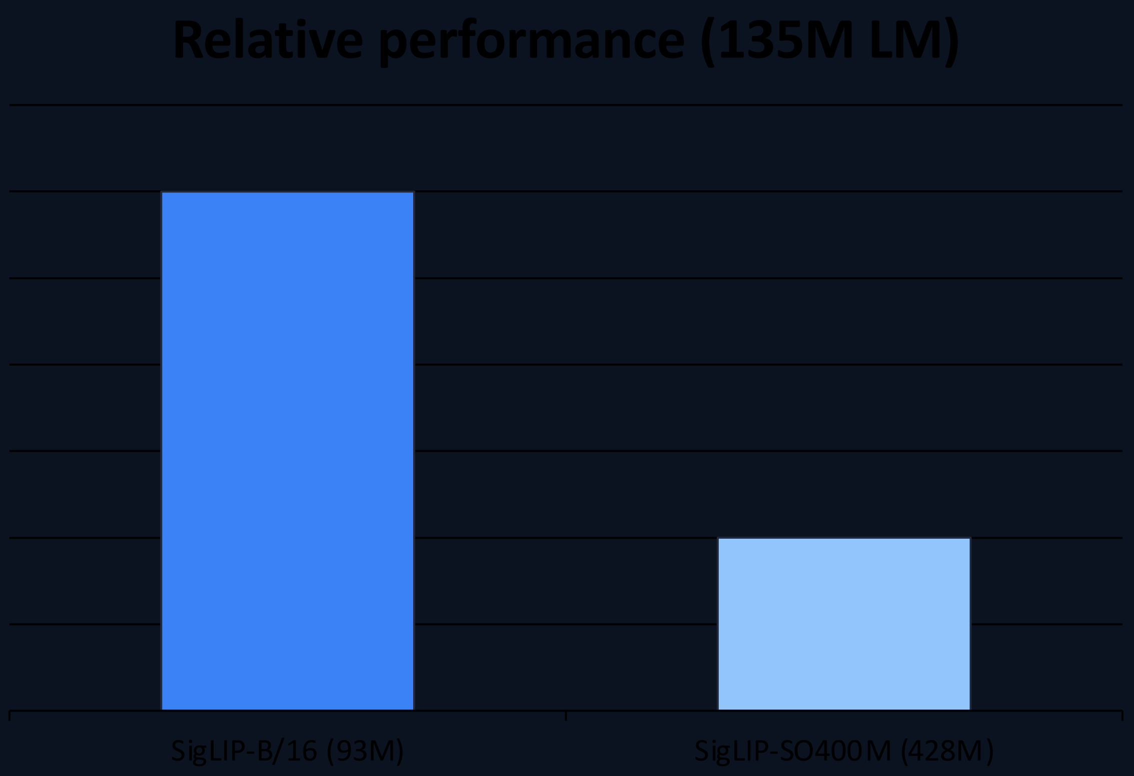
SmolVLM Architecture: Image Splitting, Pixel Shuffle, and Smol



Pipeline: split image → SigLIP encoder → pixel shuffle compression → linear projection → concatenate with text tokens → SmolLM2.

Video: sample frames and reuse the same visual-token pathway.

Smaller Vision Encoders Outperform Larger Ones in Compact V



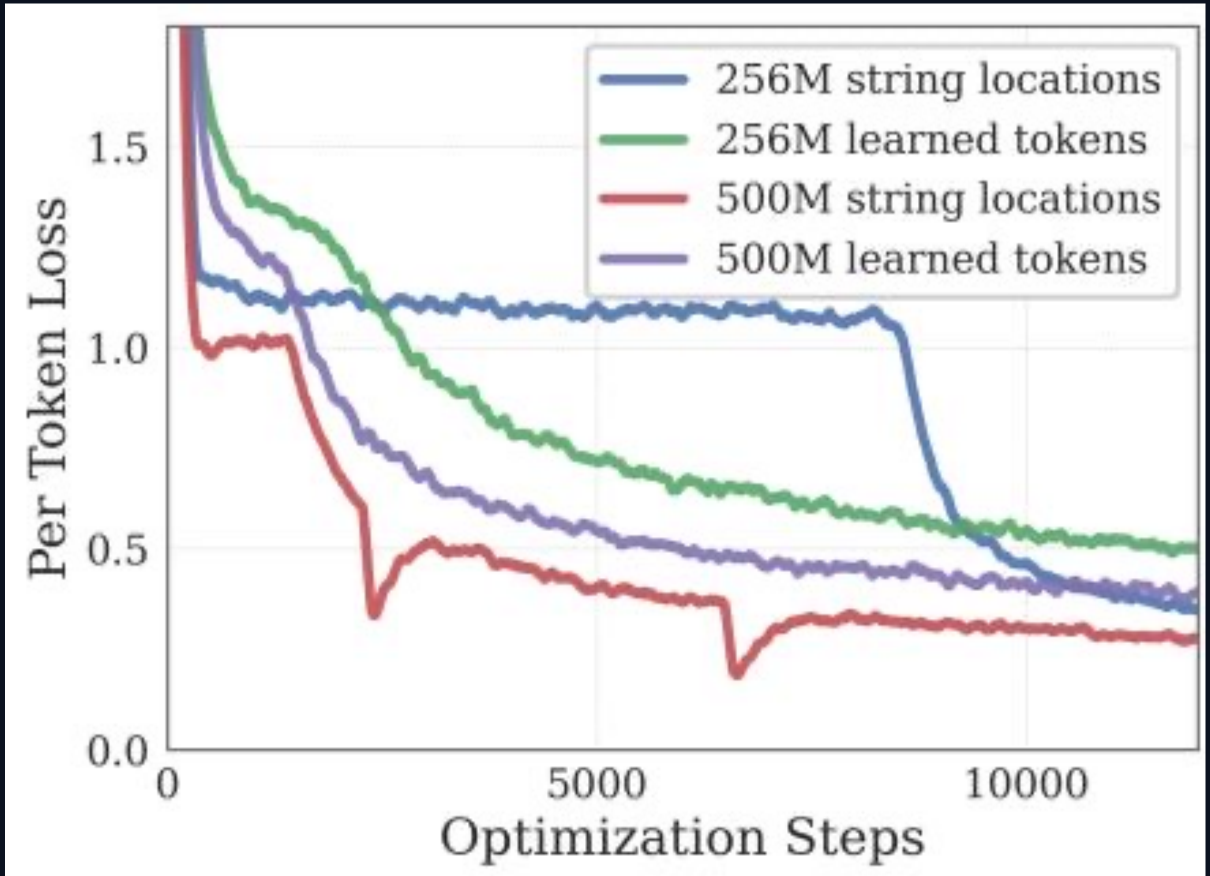
Finding 1: Balanced encoder–LM compute matters in compact VLMs.

93M SigLIP-B/16 + 135M LM outperforms 428M SigLIP-SO400M + 135M LM.

Larger encoder becomes worthwhile only at ~1.7B LM scale (≈10% parameter overhead).

Qualitative bar heights; result from paper finding.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small



Grid features → tokens

Pixel shuffle (r=4)
space→depth

16× fewer visual tokens
(r^2 reduction)

Finding 3: Small VLMs benefit from r=4 compression.

Larger models typically prefer r=2.

Extending Context from 2K to 16K Tokens Unlocks Higher Image

Context extension

2K → 16K tokens improves performance significantly

How it's done

RoPE base: 10K → 273K

Fine-tune on mixed long/short-context data

Model settings

SmolVLM-2.2B: 16K context

SmolVLM-256M/500M: 8K context

Enables higher effective image resolution under a fixed token budget

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9

Model	Params	Avg score (13)	RAM (GB, b=1)
SmolVLM-256M	256M	44.0%	0.8
SmolVLM-500M	500M	51.0%	1.2
SmolVLM-2.2B	2.2B	59.8%	4.9
Molmo-A1B-7B	7B	—	27.7
InternVL2-2B	2B	—	—

SmolVLM-256M runs with <1GB RAM and outperforms the 300× larger Idefics-80B.

Best Efficient OS baselines: Molmo-A1B-7B (27.7GB RAM) and InternVL2-2B (video).

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Ap

On-device via HuggingSnap (iOS): fully local photo/video processing (no cloud).

SmolVLM-256M reaches up to ~80 tokens/sec on Apple Silicon (reported).

Throughput scales roughly linearly with batch size for the smallest model.

Key Takeaways: Design Principles for Efficient Multimodal Mode

- Balance encoder–LM compute: smaller vision encoders can win at small LM scale.
- Use aggressive token compression for small models: pixel shuffle r=4 gives 16× fewer visual tokens.
- Scale context length to unlock higher resolution: extend 2K → 16K via RoPE base 10K → 273K + mixed-context tuning.
- Sub-1GB inference is achievable: SmolVLM-256M uses 0.8GB RAM (batch=1) with competitive accuracy.
- Fully open-sourced: weights, datasets, code, and a mobile app (HuggingSnap).